LEARNER GUIDE

Angles in the same segment

Circle Theorems · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A chord AB cuts a circle into two **segments**. Choose any point on the arc of one segment — P, or Q, or any other point on that same arc — and join it to A and B. You always get the **same angle**. The reason is the keystone once more: each of those angles at the circumference is **half the one angle at the centre**, ∠AOB. One central angle means one shared half, so every angle standing on that arc is equal. The theorem: **angles in the same segment are equal**. (If two points sit on *opposite* sides of the chord, in different segments, the angles are no longer equal — they add to 180°, which is the next lesson.)

What you are learning to notice

- **Angles in the same segment are equal** — two points on the same arc see a chord at the same angle.
- It is the keystone again: each angle at the circumference is half the same central angle.
- Look for a shared **chord** and the **same arc** — that is the signal to use this theorem.
- Same side of the chord = equal; opposite sides = supplementary (they add to 180°).

Moving through it, one step at a time

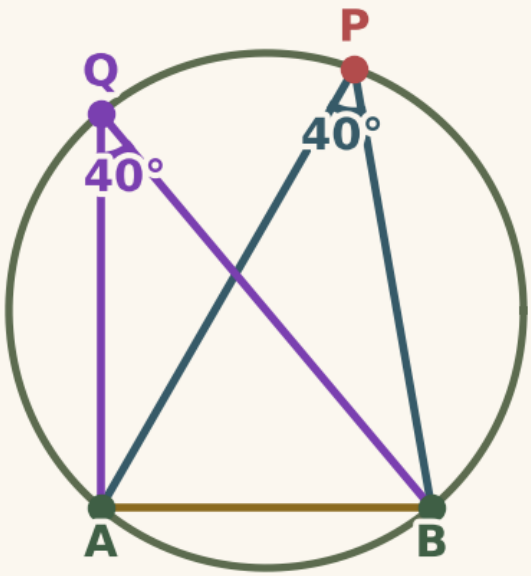
Spot two angles at the circumference that stand on the same chord AB from the same arc (the same side of AB). They are equal — write the value with the reason 'angles in the same segment'. If you are given the angle at the centre instead, halve it to get any angle in that segment.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Equal angles in the same segment

P and Q are on the same arc of a circle. ∠APB = 40°. Find ∠AQB.

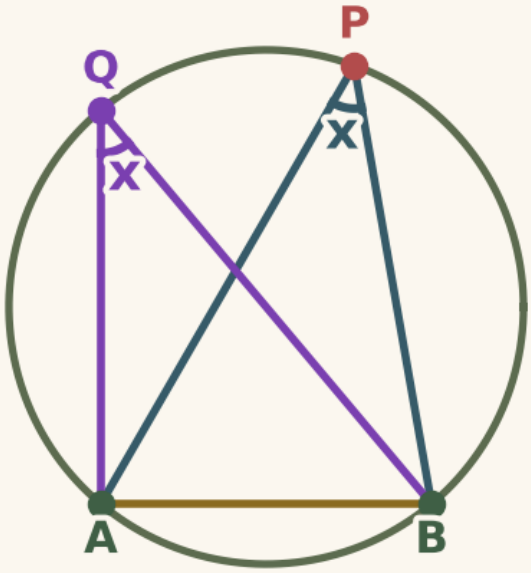


1. P and Q stand on the same arc AB (same side of the chord).	same segment
2. Angles in the same segment are equal.	∠AQB = ∠APB

∠AQB = 40°, because angles in the same segment are equal (each is half the same central angle).

Example 2 — Why they are equal

Prove ∠APB = ∠AQB for P and Q on the same arc. Let the angle at the centre be ∠AOB = 2x.



1. Draw the angle at the centre ∠AOB, and call it 2x.	∠AOB = 2x
2. By the keystone, ∠APB = ½ × 2x = x.	∠APB = x
3. ∠AQB is half the SAME central angle, so ∠AQB = x too.	∠AQB = x

∠APB = ∠AQB = x. Both are half the one central angle 2x, so they are equal — angles in the same segment. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the angle at the centre theorem (Lesson 2), the idea of an angle standing on an arc, and what a segment of a circle is (Lesson 1)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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